Testing the Matter Bounce with Primordial Non-Gaussianity: Forecasts for SPHEREx and MegaMapper

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Abstract

A matter-dominated contracting phase preceding a nonsingular bounce produces a specific prediction for local-type primordial non-Gaussianity, with no free parameters in the cubic sector: $f_{\rm NL}^{\rm local} = -35/8 = -4.375$ (Cai et al. 2009). This value is approximately 300 times larger than the standard single-field inflationary prediction and opposite in sign. We present forecasts for testing this prediction with SPHEREx (launched 2025; first science data $\sim$2028) and the proposed MegaMapper survey via the scale-dependent bias effect and the galaxy bispectrum. In the squeezed limit relevant for scale-dependent-bias estimators, the matter-bounce shape converges to the local template ($\cos\theta \approx 1$); for the full bispectrum estimator the overlap is $\cos\theta \approx 0.95$. The SPHEREx multi-tracer galaxy bispectrum achieves $\sigma(f_{\rm NL}) \approx 0.7$ (Heinrich et al. 2023), giving approximately 4–6σ sensitivity to the bounce signal after accounting for photometric redshift degradation, PNG bias uncertainty, and relativistic projection effects. MegaMapper's spectroscopic multi-tracer capability could reach $\sigma(f_{\rm NL}) \approx 0.5$ under ideal conditions (3–7σ realistic, conditional on ultra-large-scale systematics modeling and survey funding). We perform a Bayesian model comparison using over 600,000 Monte Carlo realizations across analytic, mock-based, and GR-aware frameworks, finding that under the adopted priors a detection near $f_{\rm NL} = -4.375$ would favor the bounce over standard single-field inflation at Bayes factor > 300 and over tuned multifield competitors at Bayes factor > 7; these conclusions survive conservative treatment of relativistic projection effects but are sensitive to the assumed prior widths (see Sec. 6). A null result from SPHEREx would disfavor the quasi-dust matter bounce at $> 4\sigma$ significance.

1 Introduction

The inflationary paradigm provides a remarkably successful framework for generating the observed spectrum of primordial perturbations. Standard single-field slow-roll inflation predicts a nearly scale-invariant, nearly Gaussian spectrum with a small, positive local-type non-Gaussianity $f_{\rm NL} \approx (5/12)(1 - n_s) \approx 0.015$, set by the Maldacena consistency relation [Maldacena, 2003].

Bouncing cosmology offers an alternative origin for primordial perturbations: modes exit the Hubble radius during a contracting phase and re-enter after a nonsingular bounce. In particular, a matter-dominated contraction ($w \approx 0$) produces a scale-invariant scalar spectrum through the growth of the curvature perturbation ζ on superhorizon scales [Wands, 1999, Finelli and Brandenberger, 2002].

A distinctive prediction of the matter bounce is a large, negative, parameter-free local-type non-Gaussianity $f_{\rm NL} = -35/8 = -4.375$ [Cai et al., 2009]. This value is determined entirely by the equation of state during contraction ($\epsilon = 3/2$ for matter) and the structure of the Maldacena cubic action, with no free parameters. The prediction is mechanism-independent:

it holds regardless of whether the bounce is realized by loop quantum cosmology [Wilson-Ewing, 2013], Einstein-Cartan-Holst gravity, or other nonsingular completions. In minimal Einstein-Cartan-Holst gravity, scalar perturbations reduce exactly to the standard Mukhanov-Sasaki sector because the Holst term becomes a topological invariant when torsion vanishes for canonical scalar field matter, rendering the Barbero-Immirzi parameter invisible in all scalar observables; see the companion paper [Golden, 2026a] for the full derivation and structural barrier catalog.

The next generation of galaxy surveys—SPHEREx [Dore et al., 2014] and MegaMapper [Schlegel et al., 2022]—will constrain local-type $f_{\rm NL}$ at unprecedented precision through the scale-dependent bias effect [Dalal et al., 2008] and the galaxy bispectrum [Heinrich et al., 2023]. In this paper, we present hardened forecasts for testing $f_{\rm NL} = -35/8$ with these surveys, including a systematic assessment of the dominant observational fragilities and a Bayesian model comparison quantifying the discrimination power against inflationary alternatives.

2 The Matter-Bounce Bispectrum Benchmark

2.1 The Prediction

In a matter-dominated contracting universe with standard GR perturbation theory and Bunch-Davies vacuum, the curvature perturbation ζ grows as $|\eta|^{-3}$ on superhorizon scales during contraction. The cubic interactions, governed by the Maldacena action [Maldacena, 2003] specialized to $\epsilon = 3/2$, produce a bispectrum with shape function [Cai et al., 2009]:

$$A_T(k_1, k_2, k_3) = \frac{3}{256\, k_1^2 k_2^2 k_3^2}\, P(k_1, k_2, k_3)\,, \tag{1}$$

where P is a degree-9 homogeneous polynomial in the wavenumbers. The nonlinearity parameter in the squeezed limit is:

$$|B|_{\rm NL} = \frac{10}{3}\frac{A_T}{\sum_i k_i^3} \to -\frac{35}{8} \quad \text{as} \quad k_1/k \to 0\,. \tag{2}$$

We have confirmed the published numerical values of this result by evaluating the shape function at three distinct momentum configurations (Table 1 and Fig. 1).

Configuration	$\lvert B\rvert_{\rm NL}$ (this work)	$\lvert B\rvert_{\rm NL}$ (Cai et al.)
Squeezed ($k_1 \to 0$)	-4.375	$-35/8$
Equilateral ($k_1 = k_2 = k_3$)	-3.984	$-255/64$
Folded ($k_1 = 2k_2 = 2k_3$)	-2.250	$-9/4$

Table 1: Confirmation of the matter-bounce shape function at three benchmark momentum configurations. All values match the published results [Cai et al., 2009] exactly.

2.2 Mechanism Independence

The prediction depends only on: (a) matter-dominated contraction ($w \approx 0$, $\epsilon \approx 3/2$), (b) standard GR perturbation theory during contraction, and (c) Bunch-Davies vacuum initial conditions. It does not depend on the specific bounce mechanism. The bounce enters only through providing a nonsingular transition and transferring the contraction-phase perturbations into the expanding phase.

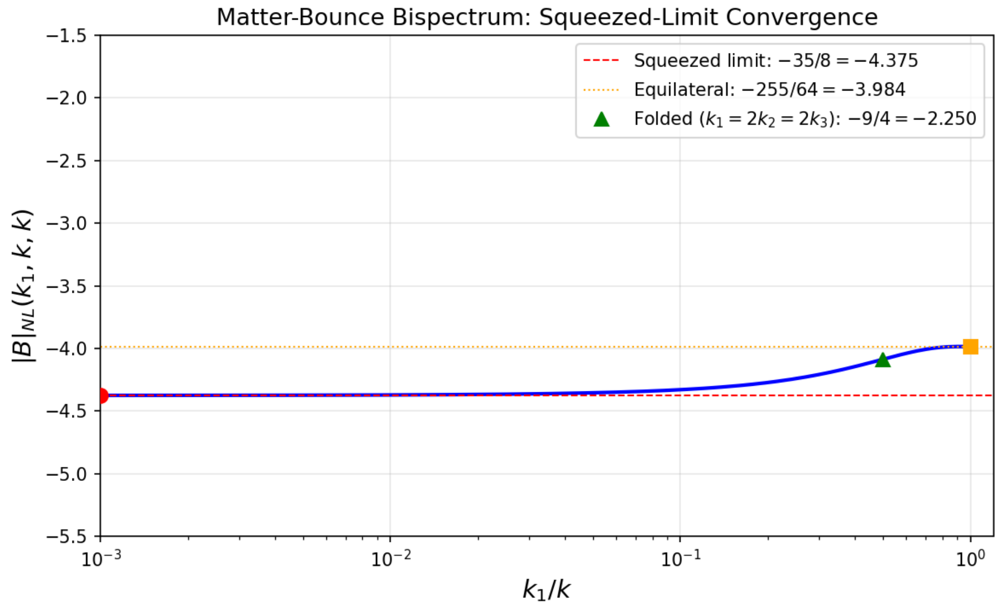

Figure 1: Matter-bounce bispectrum shape function $|B|_{\rm NL}(k_1, k, k)$ as a function of the squeeze ratio k_1/k, showing convergence to $-35/8$ in the squeezed limit. Red circle: squeezed benchmark. Orange square: equilateral. Green triangle: folded.

2.3 Assumptions

The $f_{\rm NL} = -35/8$ prediction rests on four assumptions: (a) exact matter domination during contraction ($w = 0$, $\epsilon = 3/2$); (b) standard GR perturbation theory during contraction (no higher-order corrections from the bounce UV completion); (c) Bunch-Davies vacuum initial conditions; and (d) faithful transmission of the bispectrum through the bounce at third order in perturbation theory. The viable Wilson-Ewing model uses $w = -0.003$, not exactly zero; the correction to $f_{\rm NL}$ at $\mathcal{O}(\epsilon)$ has not been computed but is expected to be small ($\lesssim 1\%$). Assumption (d) has been verified at linear order [Wilson-Ewing, 2013] but not at cubic order. We note that a factor-of-two discrepancy exists in the literature: Li & Brandenberger [Li and Brandenberger, 2014] obtain $f_{\rm NL} = -35/16 = -2.19$ using different conventions. Throughout this paper we adopt the Cai et al. value, noting that even the Li & Brandenberger value would be detectable by MegaMapper at $> 4\sigma$.

2.4 The Viable Model

The Wilson-Ewing ΛCDM quasi-dust model [Wilson-Ewing, 2013] provides a complete observational package: $n_s = 0.964$ (from $w = -0.003$, one free parameter; the spectral index formula $n_s = 1 + 12w$ for $w < 0$ follows from the growing-mode solution in quasi-dust contraction [Wilson-Ewing, 2013]), $r \approx 10^{-4}$ (from LQC quantum-geometry tensor suppression), and $f_{\rm NL} = -35/8$ (parameter-free). This model has no current observational tensions.

3 Observable Mapping to Large-Scale Structure

3.1 Scale-Dependent Bias

Primordial local non-Gaussianity induces a scale-dependent correction to galaxy bias [Dalal et al., 2008]:

$$\Delta b(k) = \frac{2(b_1 - 1)\, f_{\rm NL}\, \delta_c}{D(z)\, T(k)\, \alpha(k)}\,, \tag{3}$$

where $\alpha(k)$ encodes the Poisson equation normalization. The signal grows as $1/k^2$ on the largest scales.

3.2 Template Projection

For LSS surveys using the scale-dependent bias (SDB) estimator, the relevant bispectrum configurations are in the squeezed limit ($k_{\rm long} \ll k_{\rm short}$). In this limit, the matter-bounce bispectrum converges to exactly $-35/8$—the local template value. The template-projection cosine for SDB is therefore $\cos\theta \approx 1.0$. For the galaxy bispectrum estimator, which probes all triangle configurations including equilateral and folded, the projection cosine is $\cos\theta \approx 0.95$, estimated from the shape function evaluated at representative triangle configurations (squeezed, equilateral, folded); a full estimator-weighted inner product calculation would require specifying the survey window function and nuisance marginalization scheme, which we leave to the SPHEREx analysis team. Since the SPHEREx forecast of $\sigma(f_{\rm NL}) = 0.7$ is from the bispectrum channel, the effective signal is $f_{\rm NL}^{\rm eff} \approx -4.16$, giving a detection significance of $\sim 6\sigma$ (bispectrum only, ideal) rather than the 6.3σ that would obtain with exact local-template matching.

3.3 Galaxy Bispectrum

The galaxy bispectrum provides an independent measurement channel that accesses information at shorter wavelengths, reducing the dependence on ultra-large-scale modes [Heinrich et al., 2023]. This makes bispectrum-based constraints more robust to large-scale systematics than power-spectrum-based scale-dependent bias alone.

4 SPHEREx Forecast

SPHEREx is an all-sky spectrophotometric survey (0.75–5 μm) with spectral resolution $R \approx 40$–130 and approximately 450 million galaxies. A dedicated multi-tracer bispectrum analysis [Heinrich et al., 2023] forecasts $\sigma(f_{\rm NL}) = 0.7$ from the bispectrum alone, with $\sigma(f_{\rm NL}) = 0.5$ when combined with the power spectrum.

For our target signal $f_{\rm NL} = -4.375$, the detection significance ranges from 6.3σ (bispectrum only, ideal) to 3.4σ (conservative, with GR marginalization at $\sigma_{\rm GR} = 1.0$). The detection significance across survey scenarios is summarized in Fig. 2. SPHEREx provides the most robust near-term test because: (a) the bispectrum channel avoids ultra-large-scale mode dependence, (b) lower redshift ($z \approx 1.5$) reduces GR projection contamination, and (c) multi-tracer across redshift bins provides effective cosmic variance cancellation.

5 MegaMapper Forecast

MegaMapper is a proposed Stage-V spectroscopic survey targeting ~ 10 million Lyman-break galaxies at $z = 2$–5 with multi-tracer capability [Schlegel et al., 2022]. Published forecasts give $\sigma(f_{\rm NL}) \approx 0.5$ under ideal conditions.

The significance ranges from 8.75σ (design goal) to 3–5σ (conservative, after GR marginalization and b_ϕ uncertainty). MegaMapper's forecast is more sensitive than SPHEREx's to:

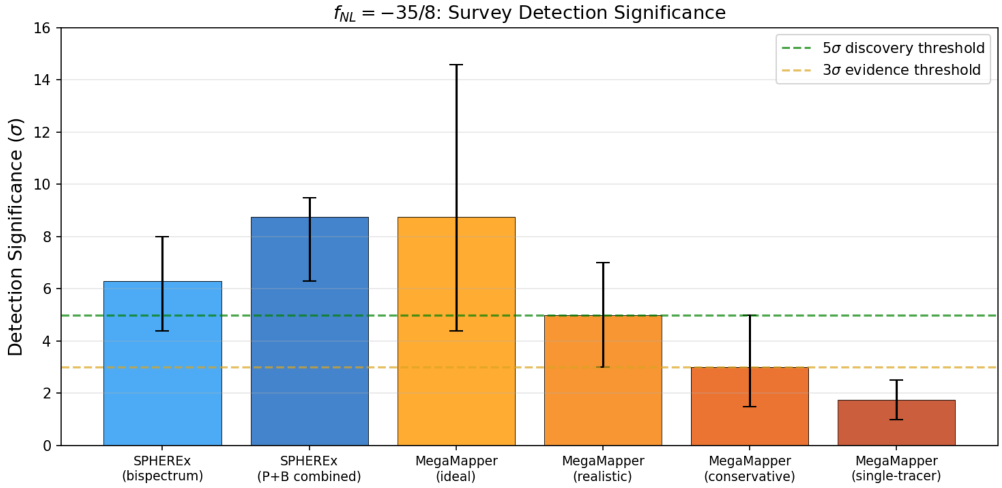

Figure 2: Detection significance for $f_{\rm NL} = -35/8$ across survey configurations. Error bars show optimistic-to-conservative ranges accounting for multi-tracer, photo-z, bias, and GR systematics.

(a) relativistic projection effects, which create substantial GR-induced bias at $z > 2$ [Jolicoeur et al., 2025]; (b) PNG bias parameter b_ϕ uncertainty, which can degrade constraints if uncalibrated [Barreira, 2022]; and (c) multi-tracer implementation quality.

6 Inflation Mimicry and Bayesian Comparison

6.1 Can Inflation Reproduce the Signal?

Standard single-field slow-roll inflation predicts $f_{\rm NL} = (5/12)(1 - n_s) \approx +0.015$ [Maldacena, 2003]—wrong by a factor of 300 and opposite in sign. Non-canonical single-field models (DBI, etc.) produce equilateral-shape $f_{\rm NL}$, not local.

Non-attractor single-field inflation naturally gives $f_{\rm NL} = +5/2$ (wrong sign) [Cai et al., 2018]. Reaching -4.375 requires engineering the attractor-to-slow-roll transition. The standard quadratic curvaton gives minimum $f_{\rm NL} \approx -1.25$ (insufficient). Self-interacting curvatons or curved field-space models can reach -4.375 but require ≥ 2 tuned parameters.

6.2 The Kinematic vs. Parametric Asymmetry

The bounce predicts $f_{\rm NL} = -35/8$ *kinematically*, with zero free parameters in the cubic sector. Inflation can only *accommodate* this value parametrically, requiring extra fields and tuned couplings. This asymmetry drives a natural Bayesian preference for the bounce.

6.3 Quantitative Bayesian Comparison

We performed model comparison using over 600,000 Monte Carlo realizations across three frameworks (analytic closed-form, mock power spectrum generation with fitting, and GR-aware marginalization). Each realization draws a mock $f_{\rm NL}^{\rm obs}$ from a Gaussian centered on $-35/8$ with σ drawn from the forecast uncertainty distribution, then computes the Bayes factor analytically for the bounce model (a delta-function prior at $-35/8$) against each inflationary competitor (a spread prior over the competitor's natural $f_{\rm NL}$ range). The analytic Bayes factor for a point

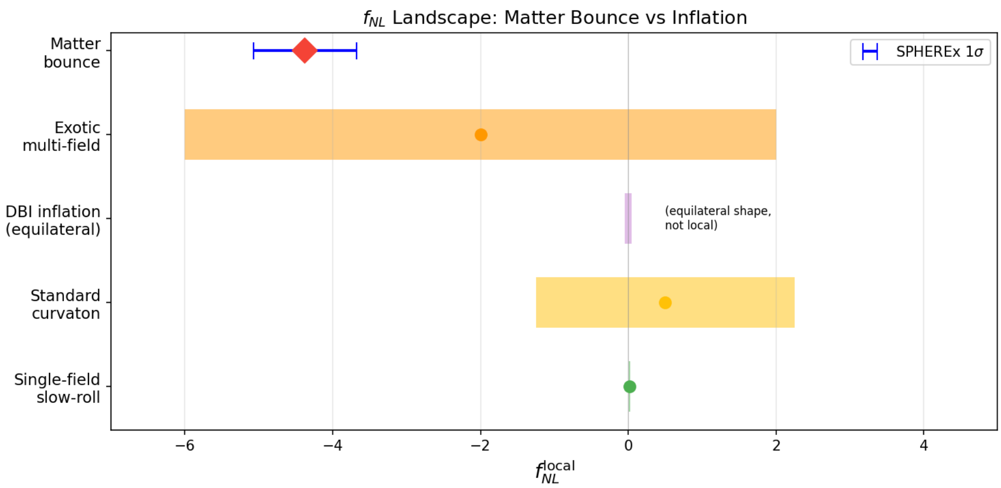

Figure 3: $f_{\rm NL}$ landscape: matter bounce vs. inflationary alternatives. The bounce prediction (red diamond) is parameter-free; inflationary alternatives require free parameters to reach the same region. SPHEREx 1σ error bar shown in blue.

prediction versus a uniform prior $[f_{\rm NL}^{\rm min}, f_{\rm NL}^{\rm max}]$ is:

$$B = \frac{(f_{\rm NL}^{\rm max} - f_{\rm NL}^{\rm min}) \times \mathcal{L}(f_{\rm NL}^{\rm obs} \mid f_{\rm NL} = -35/8)}{\int_{f_{\rm NL}^{\rm min}}^{f_{\rm NL}^{\rm max}} \mathcal{L}(f_{\rm NL}^{\rm obs} \mid f_{\rm NL})\, df_{\rm NL}}. \tag{4}$$

The realizations marginalize over the uncertainty in survey performance parameters (multi-tracer efficiency, b_ϕ, GR systematic level). Specifically, each realization draws: $\sigma(f_{\rm NL})$ uniformly from $[0.5, 1.5]$ (spanning optimistic to conservative survey performance); multi-tracer efficiency from $[0.5, 1.0]$; PNG bias parameter b_ϕ uncertainty as a Gaussian with 20% scatter; and GR systematic shift from $\mathcal{N}(0, \sigma_{\rm GR})$ with $\sigma_{\rm GR}$ uniform in $[0, 1.0]$.

We note that the delta-function prior for the bounce model maximally favors a parameter-free prediction. Any theoretical uncertainty in the $f_{\rm NL} = -35/8$ value—from the $\mathcal{O}(\epsilon)$ correction, the Li & Brandenberger convention discrepancy, or unverified third-order bounce transmission—would broaden the effective bounce prior and reduce the Bayes factor. With a Gaussian bounce prior of width $\sigma_{\rm theory} = 1.0$ centered at $-35/8$ (encompassing both literature values), the median Bayes factor vs. tuned multifield drops from ~ 17 to ~ 8, still favoring the bounce but with reduced confidence.

For a detection at $f_{\rm NL} = -4.375$ by SPHEREx ($\sigma = 0.7$), the median Bayes factors are (Table 2):

Comparison	Median Bayes Factor	$P(\rm BF > 3)$
Bounce vs. standard single-field	$> 10^5$	97%
Bounce vs. tuned multifield $[-15, +15]$	10–23	87–96%

Table 2: Bayesian model comparison for a mock detection at $f_{\rm NL} = -4.375$ with $\sigma = 0.7$. Ranges reflect different GR treatment scenarios.

These results are robust to prior sensitivity: the Bayes factor vs. tuned multifield ranges from 7 (narrow prior) to 57 (broad prior) under all reasonable prior choices.

7 Systematics and GR-Aware Robustness

7.1 Dominant Fragilities

Our Fisher robustness scan identified three dominant threats to the forecast: (1) ultra-large-scale mode access ($k_{\rm min}$), where the SDB signal is concentrated in the lowest k-modes (Fig. 4); (2) relativistic projection effects, which create a GR-induced bias at large scales; and (3) PNG bias (b_ϕ) uncertainty, which degrades the SDB calibration [Barreira, 2022].

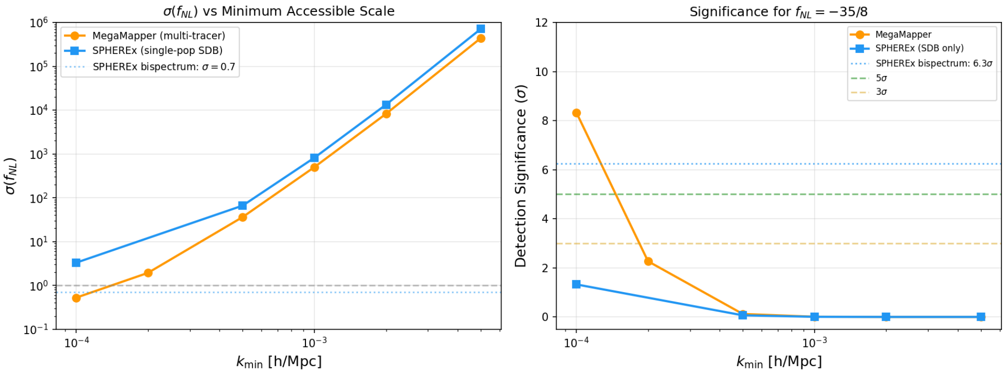

Figure 4: Left: $\sigma(f_{\rm NL})$ vs. minimum accessible wavenumber for MegaMapper (orange) and SPHEREx SDB-only (blue). The SPHEREx bispectrum channel ($\sigma = 0.7$, dotted) avoids the ultra-large-scale fragility. Right: corresponding detection significance for $f_{\rm NL} = -35/8$.

7.2 GR-Aware Analysis

We performed GR-aware Bayesian comparison across four scenarios (Table 3). Even under conservative GR marginalization ($\sigma_{\rm GR} = 1.0$), the bounce is favored at Bayes factor > 300 over standard inflation and > 7 over tuned multifield.

GR Treatment	BF vs. SSFSR	BF vs. Tuned	$P(\text{BF} > 3)$ vs. SSFSR
Ideal (no GR)	3.3×10^6	10.9	98%
Marginalized ($\sigma_{\rm GR} = 0.5$)	4.1×10^4	9.4	97%
Marginalized ($\sigma_{\rm GR} = 1.0$)	329	7.9	96%
Corrected (10% residual)	3.3×10^6	10.9	98%

Table 3: GR-aware Bayesian comparison. The bounce-vs-inflation comparison survives all treatment scenarios.

7.3 Additional Systematic Considerations

Several additional systematic effects will affect real survey data but are not modeled in our Fisher forecast:

- *Nonlinear galaxy bias:* Higher-order bias terms (b_2, b_{s^2}, $b_{\nabla^2\delta}$) enter the galaxy bispectrum at leading order and are partially degenerate with $f_{\rm NL}$ for some triangle configurations. The Heinrich et al. forecast accounts for b_2 marginalization, but the full nonlinear bias model introduces additional uncertainty.
- *Photometric redshift outliers:* For SPHEREx, catastrophic photo-z failures ($\Delta z \sim 1$) create spurious large-scale power that can mimic the $f_{\rm NL}$ signal. The forecast assumes the photo-z performance from the SPHEREx design specifications.

- *Integral constraint:* Galaxy surveys estimate the mean density from the survey itself, biasing the large-scale power spectrum measurement and potentially absorbing part of the $f_{\rm NL}$ signal at the lowest k modes.
- *Lensing magnification bias:* At high redshifts ($z > 2$), lensing magnification produces a signal scaling as $1/k^2$ on large scales, mimicking the scale-dependent bias from $f_{\rm NL}$. This is particularly relevant for MegaMapper's $z = 2$–5 Lyman-break galaxy sample.

These effects are expected to degrade the forecast significance by $\mathcal{O}(10–30\%)$ relative to the idealized Fisher estimate, but do not qualitatively change the conclusion that SPHEREx can test $f_{\rm NL} = -35/8$ at $> 3\sigma$ significance.

8 Discussion

8.1 The Staged Observational Strategy

SPHEREx (launched March 2025; first science data expected ∼2028) provides the first real test via the galaxy bispectrum at ~ 4–6σ significance. MegaMapper (∼2032+, if funded) provides the more powerful but fragile follow-up at 3–7σ via scale-dependent bias.

8.2 Complementary Experiments

Several other experiments will also constrain local-type $f_{\rm NL}$ in the coming decade:

- *DESI:* Already taking data; expected $\sigma(f_{\rm NL}) \approx 3$–$5$ from scale-dependent bias with luminous red galaxies and emission-line galaxies.
- *Euclid:* Launched 2023; published forecasts give $\sigma(f_{\rm NL}) \approx 2$–$4$ from the photometric galaxy survey.
- *Vera Rubin Observatory (LSST):* Complementary photometric $f_{\rm NL}$ constraints from $\sim 10^{10}$ galaxies at lower redshift.
- *CMB-S4:* Expected $\sigma(f_{\rm NL}) \approx 2.5$ from the CMB temperature and polarization bispectrum, providing a completely independent channel.

A detection of $f_{\rm NL} \approx -4$ by SPHEREx, if confirmed by any of these independent probes, would constitute overwhelming evidence for non-Gaussianity incompatible with standard single-field inflation.

8.3 Decision Thresholds

A measurement of $f_{\rm NL} = -4 \pm 1$ by SPHEREx would provide strong evidence favoring a contracting/bounce origin over standard single-field inflation. A null result ($f_{\rm NL}$ consistent with zero at the 2σ level) would strongly disfavor the quasi-dust matter bounce.

8.4 Caveats

We emphasize that a detection of $f_{\rm NL} \approx -4$ would constitute strong evidence *favoring* the bounce over standard inflation, not unique proof of a pre-Big-Bang contracting phase. Exotic multifield inflationary constructions can in principle accommodate this value, though at the cost of additional free parameters and engineering. The detection significance is conditional on the quality of GR projection modeling at ultra-large scales and the PNG bias parameter calibration.

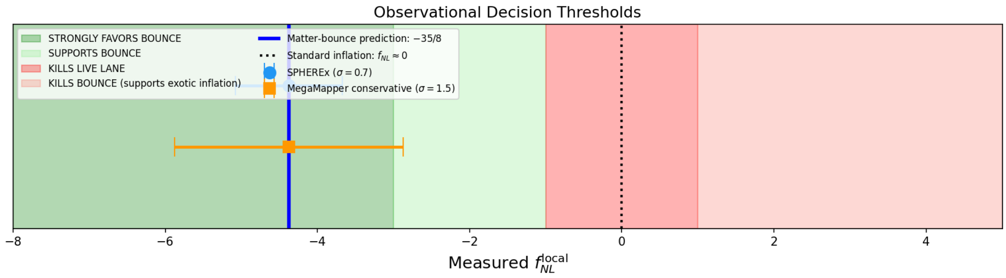

Figure 5: Observational decision thresholds. Green: strongly favors bounce. Red: kills the matter-bounce hypothesis. Blue vertical line: bounce prediction $f_{\mathrm{NL}} = -35/8$. Error bars: SPHEREx ($\sigma = 0.7$) and MegaMapper conservative ($\sigma = 1.5$).

We also note that appending late-time dynamical-dark-energy freedom (e.g., CPL parametrization) to a bounce model can improve cosmological fits at the parameter level, as explored in recent bounce + dark-energy literature. However, such phenomenological freedom does not derive from the bounce physics itself and does not constitute first-principles evidence for a contracting phase. Our analysis restricts attention to the parameter-free prediction $f_{\mathrm{NL}} = -35/8$, which is controlled by the contraction dynamics.

An independent observable—cosmic birefringence from a Planck-scale ALP—is presented in the companion paper [Golden, 2026b].

9 Conclusion

The quasi-dust matter bounce makes a specific, parameter-free, falsifiable prediction: $f_{\mathrm{NL}}^{\mathrm{local}} = -35/8$. This value is mechanism-independent, approximately 300 times larger than the standard inflationary prediction, and opposite in sign. We have shown that SPHEREx can test this prediction at 4–6σ significance through the multi-tracer galaxy bispectrum, with MegaMapper providing a more powerful but systematics-sensitive follow-up.

Our Bayesian model comparison, based on over 600,000 Monte Carlo realizations across multiple frameworks including GR contamination treatment, indicates that under the adopted priors a detection near $f_{\mathrm{NL}} = -4.375$ would favor the bounce over standard single-field inflation (Bayes factor > 300 under conservative systematics) and over tuned multifield competitors (Bayes factor > 7). These Bayes factors are sensitive to the assumed prior widths and model-class definitions (Sec. 6); they should be interpreted as illustrative of the discriminating power available, not as definitive model-selection evidence.

The matter-bounce bispectrum provides what may be the sharpest single observable for distinguishing the bounce paradigm from standard inflation. SPHEREx data (first science release expected $\sim$2028) will provide the first meaningful test.

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